

CoRoT-2b

R-2284

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_110815 · created 2026-08-17T00:21:07+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455788.8027 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.181 +/- 0.065
Transit depth (Rp/Rs)^2	3.26 +/- 2.33 [%]
Orbital Inclination (inc)	89.37 +/- 2.03
Airmass coefficient 1 (a1)	1.3 +/- 0.3
Airmass coefficient 2 (a2)	-0.12 +/- 0.12
Scatter in the residuals of the lightcurve fit is	23.28 %
Best Comparison Star	#1 - [151, 395]
Optimal Aperture	4.15
Optimal Annulus	10.37
Transit Duration (day)	0.055 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.181 ± 0.065 vs 0.1667 (deviation 0.18σ)
Duration fit vs published	0.055 d vs 0.095 d (deviation 9.41σ)
Mid-time O–C	-1.6 min
Depth SNR	2.3
Residual scatter	23.28 %

Light curve

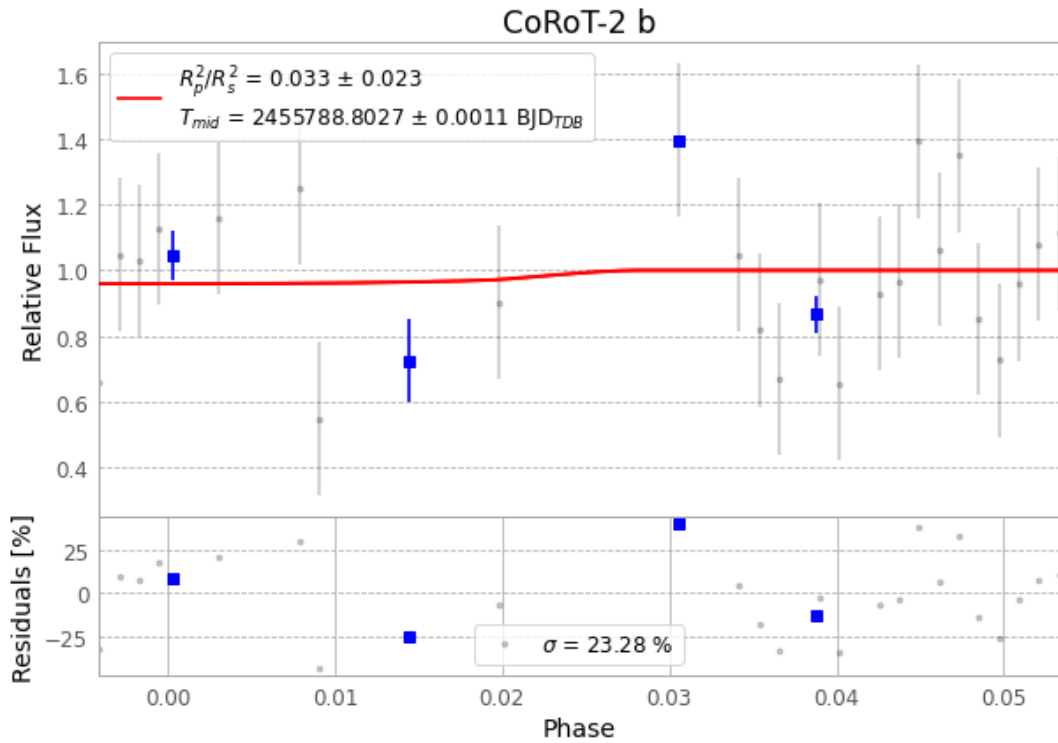


Figure 1 — FinalLightCurve_CoRoT-2 b_2011-08-14.png (SHA-256 b346bc4dabf30cd0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [205, 368], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8e7a8f703ea4f8dc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

49 FITS frames (32 MB), CoRoT-2110815065712.FITS → CoRoT-2110815092112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-110815.log (44081c8c9e9e...), FinalLightCurve_CoRoT-2 b_2011-08-14.png (b346bc4dabf3...), FinalLightCurve_CoRoT-2 b_2011-08-14.csv (284d6c393d0c...)

Compute resources

Wall time 159.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)